VS-08 Innovation and Methodological Development

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VS-08 Innovation and Methodological Development

1. Purpose and value created

VS-08 makes sure the statistical organisation does not stand still. It discovers, tests, and introduces new ways of working: new data sources, methods, tools, and technologies. It moves the organisation from "we have always done it this way" to "we can do this better, faster, and smarter." It connects research, methodology, IT, and business users. The value created is **continuous improvement**: the SAF stays future-proof for new data (IoT, administrative data, data spaces), new techniques (machine learning, small-area estimation), and new operating models (secure processing environments, data sharing).

VS-08 does not transform data. It improves the methods and tools that the primary value streams use, and its output is an enabling capability: validated methods, documented tools, reference implementations, and trained staff.

> **i** Signature discipline: controlled innovation
>
> Innovation in VS-08 is controlled. Only methods that are sound, documented, and compliant reach production. The evaluation gate in VS-08.3 stops attractive but unsound ideas, and the transfer gate in VS-08.5 moves a validated method into production only after VS-07 approval and on VS-09 platforms.

VS-08 corresponds to the horizontal business domain **Innovation**. Its domain processes are mainly realised by the business function FUN-4 *Methodology and Research*.

PLACEHOLDER: ArchiMate view VS-08 (a) value stream and capabilities

2. Relation to the value chain

VS-08 is not a stage in the production line. It runs alongside the chain as a continuous research and development pipeline. It draws its agenda from the needs of the primary value streams and feeds validated improvements back into them. It has no state transition, no quality gate of its own, and no registration event. It aligns to the GAMSO area **Capability Management**, with the innovation agenda set under **Strategy and Leadership**.

The improvement recommendations that every primary value stream produces in its *Continuous Improvement* sub-process (see Standard activities in every value stream), together with user needs surfaced by VS-10 and an external scan of technology, form the **innovation backlog** of VS-08. VS-08 is also an occasional consumer of States #1, #2, and #3, for the exploration of sources and the testing of new methods, always under specific authorisation.

Primary value stream	Provides	Receives
VS-01 Data Acquisition and Ingestion	New ways to connect to data spaces and new sources; new collection methods	Improvement needs; authorised access to #1 for source exploration
VS-02 Data Standardization and Integration	New mapping or classification techniques	Improvement needs; authorised access to #2 for exploration of sources and methods
VS-03 Data Editing and Enrichment	Smarter imputation and outlier detection; new editing and coding methods	Improvement needs; authorised access to #3 for testing new editing, imputation, or coding methods
VS-04 Estimation, Analysis and Statistical Modeling	New models and methods based on machine learning; small-area estimation	Improvement needs; fitness-for-purpose validation
VS-05 Statistical Product Composition	Validated methods and tools transferred to production	Improvement needs; fitness-for-purpose validation
VS-06 Dissemination and Service Delivery	New channels: APIs, linked data, interactive views	Improvement needs; fitness-for-purpose validation

PLACEHOLDER: ArchiMate view VS-08 (c) interactions with the value chain: improvement needs from VS-01 to VS-06, user needs from VS-10, and the external scan entering VS-08; validated methods and tools leaving through the VS-08.5 transfer gate to the target production stream (for example VS-03 or VS-04); compliance approval from VS-07 and sandbox and production platforms from VS-09

3. Stages

The stages form a continuous innovation pipeline: identify, research, evaluate, develop, integrate, and share. The last stage surfaces the next round of needs. The cadence is project or pipeline based, not per production cycle.

Stage	Name	Value added
VS-08.1	Identify Innovation Needs and Opportunities	Promising developments are known
VS-08.2	Conduct Research and Proof-of-Concepts	Ideas are tested in a limited scope
VS-08.3	Evaluate Methodological Soundness and Feasibility	Only sound and feasible innovations proceed
VS-08.4	Develop and Document New Methodologies or Tools	Innovations are ready for reuse

Stage	Name	Value added
VS-08.5	Integrate Innovation into Production Environments	Innovations run in production under governance
VS-08.6	Promote Knowledge Sharing and Capability Building	Staff can apply the new methods

VS-08.1 Identify Innovation Needs and Opportunities

Value created: *A prioritised view of the developments worth pursuing.*

The pipeline starts by scanning technological, methodological, and policy developments, such as AI, machine learning, new data sources, and cloud. It also takes in the improvement backlog of the chain: the recommendations of the primary value streams and the user-driven needs from VS-10. Candidates are prioritised by organisation-wide value and scalability, not by novelty (SAF-P04). The result is the prioritised innovation backlog.

Activity	Key outputs	SAF process	Other reference
Scan technological, methodological, and policy developments (AI, machine learning, new sources, cloud)	Scan results	No reference process (NSI-specific)	2.2.1 Data source exploration; 2.2.3 Methodology
Collect improvement recommendations from the primary value streams	Backlog entries (*Innovation need raised*)	PRD-8.3 Agree an action plan	4.1.4 Improvement management
Collect user needs that imply new methods, from VS-10	Backlog entries (*User-driven innovation need*)	PRD-1.1 Identify needs	2.2.5 User research
Check whether new or existing data sources can meet the need	Source assessment	PRD-1.5 Check data availability and suitability	SAF-P08
Prioritise candidates by organisation-wide value and scalability	Prioritised innovation backlog	PRD-1.6 Prepare and submit business case	SAF-P04

VS-08.2 Conduct Research and Proof-of-Concepts

Value created: *Evidence on whether an idea works, before any commitment.*

Candidates from the backlog enter controlled experiments and pilots. Each tests a new method or technology in a limited scope before any commitment is made. Experiments run in research or sandbox environments that VS-09 provides. Where they need production data, they use registered versions of States #1, #2, or #3 under specific authorisation, never the working environment of a producing domain. Subject-matter experts from the production streams bring the real problem.

Activity	Key outputs	SAF process	Other reference
Define the scope and success criteria of the experiment	Experiment plan	No reference process (NSI-specific)	2.2.3 Methodology
Request a sandbox environment from VS-09	Research environment (*Platform request*)	No reference process (NSI-specific)	VS-09
Obtain authorised access to registered data of #1, #2, or #3	Access granted for a stated purpose and period	DLM-6 Provide data for reuse	SAF-P10, SAF-P11

Activity	Key outputs	SAF process	Other reference
Run the proof-of-concept or pilot in limited scope	Experiment and pilot results	No reference process (NSI-specific)	2.2.1 Data source exploration; 2.2.2 Product innovation
Record research findings	Research findings	OVA-5 Knowledge Management	SAF-P03

VS-08.3 Evaluate Methodological Soundness and Feasibility

Value created: *Only sound and feasible innovations proceed.*

Promising results are evaluated for accuracy, scalability, and alignment with the quality and ethical frameworks of the NSI. This stage is the gate that stops attractive but unsound ideas. A healthy gate stops some proofs-of-concept, and every stop has a documented rationale. Subject-matter experts confirm fitness for purpose.

Activity	Key outputs	SAF process	Other reference
Assess accuracy against the intended use	Soundness assessment	PRD-3.6 Test statistical business process	2.1.3 Statistical quality management
Assess scalability and feasibility in production conditions	Feasibility assessment	PRD-3.6 Test statistical business process	
Check alignment with the quality and ethical frameworks	Soundness and feasibility assessment	OVA-1 Quality Management	
Validate fitness for purpose with subject-matter experts	Fitness-for-purpose validation	PRD-3.6 Test statistical business process	
Decide to proceed or stop, with a documented rationale	Gate decision record	PRD-3.6 Test statistical business process	SAF-P03

VS-08.4 Develop and Document New Methodologies or Tools

Value created: *Innovations that are reproducible and ready for reuse.*

What passes the gate is developed into standardised procedures, algorithms, or reference implementations. It is documented so that it is reproducible and ready to hand over, rather than living in a researcher's notebook. Shared and open tooling comes first, and the NSI contributes back (SAF-P01). Each artefact has a designated owner and a version.

Activity	Key outputs	SAF process	Other reference
Formalise the validated method as a standardised procedure or algorithm	Standardised method	OVA-5 Knowledge Management	2.2.3 Methodology
Build the reference implementation, documented and version-controlled	Reference implementation	PRD-3.2 Reuse or build processing and analysis components	SAF-P01; 3.2.1 Application development
Document the method and tool for reproducibility	Method documentation	OVA-5 Knowledge Management	
Assign an owner and a version to each artefact	Versioned method and tool	OVA-5 Knowledge Management	SAF-P03

VS-08.5 Integrate Innovation into Production Environments

Value created: *Innovations run in production under governance.*

This is the defining stage of VS-08: the governed transfer of a validated method or tool from pilot to production in the relevant primary value stream. The method passes the compliance assessment of VS-07 before transfer and runs on VS-09 platforms. It must fit the steady-state interfaces, so that the production streams stay loosely coupled (SAF-P07). Once transferred, the method becomes part of the design metadata of the receiving stream. This is the GAMSO idea of transferring mature results of innovation into production.

Activity	Key outputs	SAF process	Other reference
Submit the method to VS-07 for compliance assessment before production	Compliance approval (*Method compliance assessment*)	PRD-3.7 Finalise production systems	VS-07
Check interoperability with the steady-state interfaces of the target stream	Interoperability confirmation	PRD-3.4 Configure workflows	SAF-P07
Onboard the method or tool on the VS-09 production platform	Production runtime (*Production onboarding*)	PRD-3.7 Finalise production systems	VS-09
Hand over the method as design metadata to the receiving stream	Method live in a production stream (*Method transferred to production*)	PRD-3.7 Finalise production systems; OVA-2 Metadata Management	GAMSO Capability Management
Train the production staff and arrange support	Trained operating team	PRD-3.7 Finalise production systems	

VS-08.6 Promote Knowledge Sharing and Capability Building

Value created: *Staff who can adopt and sustain the new methods.*

VS-08 disseminates results, best practices, and guidelines, and grows staff competency, so that the organisation can actually adopt and sustain the new way of working. Adoption in the target streams is followed up. This in turn surfaces the next round of needs for VS-08.1.

Activity	Key outputs	SAF process	Other reference
Disseminate results, best practices, and guidelines	Disseminated guidance	OVA-5 Knowledge Management	SAF-P05
Train staff on new methods	Trained staff	OVA-5 Knowledge Management	3.1.1 Personnel management (HR)
Share methods and tools with other NSIs and open-source communities	Shared methods and contributions	OVA-5 Knowledge Management	SAF-P01; 4.1.6 Collaboration
Follow up adoption in the target streams and collect new needs	Adoption measures, backlog input	PRD-8.1 Gather evaluation inputs	4.1.4 Improvement management

4. Plan, build, operate, and improve

VS-08 is itself a capability that is planned, built, operated, and improved. Unlike a primary value stream, it produces methods and capability, not statistical data. The stages group into the cycle: VS-08.1 and

VS-08.2 find what is worth doing, VS-08.3 and VS-08.4 make it sound and real, VS-08.5 puts it to work, and VS-08.6 spreads it.

Cycle step	What happens in VS-08	Main outputs	SAF process
Plan	Set the innovation strategy, aligned to the organisational strategy, and the evaluation criteria of the VS-08.3 gate	Innovation strategy, evaluation criteria	PRD-1.6 Prepare and submit business case; OVA-1 Quality Management
Build	Build research and sandbox environments, with VS-09	Research environments	No reference process (NSI-specific)
Operate	Run the innovation pipeline continuously, from backlog to transfer	Validated methods and tools in production	PRD-3.6 Test statistical business process; PRD-3.7 Finalise production systems
Improve	Learn from VS-08.6 and from retrospectives; review the key performance indicators (attribute 9)	Improvement actions for the pipeline	PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan

Aspect	VS-08
Trigger	*Innovation need raised*; *User-driven innovation need*; results of the external scan
Cadence	Continuous pipeline; project based per innovation
Performed by	ROL-07 Methodologist, ROL-15 Subject-matter statistician (researcher or data scientist), ROL-10 Data engineer
Input	Innovation backlog; authorised access to #1, #2, #3; sandbox environments
Output	Validated methods and tools transferred to production; trained staff
Control point	Soundness and feasibility gate in VS-08.3; compliance approval by VS-07 before the transfer in VS-08.5
Exception path	Stop at VS-08.3 with a documented rationale; return to research when the compliance assessment fails

Enabling artefacts. VS-08 manages enabling artefacts, not statistical data. They are version-managed across their life cycle (SAF-P03). Each artefact has a designated owner, changes are tracked, prior versions remain identifiable, and superseded methods are formally retired. Once transferred, an artefact becomes part of the design metadata of a production stream.

Artefact	Produced in
Innovation backlog	VS-08.1
Research findings and proof-of-concept results	VS-08.2
Soundness and feasibility assessments	VS-08.3
Validated methods, algorithms, and reference implementations	VS-08.4
Knowledge assets: guidelines, training	VS-08.6

5. Actors and roles

The table shows the reference roles involved in VS-08 (see *Role names in the value streams*).

Actor or role	ID	Process type	Responsibility in VS-08
Head of methodology or innovation	ROL-01, ROL-07	Management	Process owner of VS-08; owns the innovation agenda and the methodological standards
Methodologist	ROL-07	Design	Evaluates soundness and feasibility; formalises validated methods
Researcher or data scientist	ROL-15	Execution	Conducts research and proofs-of-concept
Subject-matter expert from the production streams	ROL-15	Execution	Brings the real problem and validates fitness for purpose
Tool or platform engineer	ROL-10	Implementation	Builds reference implementations and prepares them for production, with VS-09
Platform engineer	ROL-11	Implementation	Provides sandbox and production platforms, anchored in VS-09
Compliance officer	ROL-19	Management	Approves a method against compliance criteria before transfer (VS-07)
Trainer or knowledge manager	Not in the reference set	Execution	Disseminates results and builds staff competency

6. Capabilities, processes, and functions

Capability foundations

Capability group	Capabilities	Role in VS-08
2.2 Research	2.2.1 Data source exploration, 2.2.2 Product innovation, 2.2.3 Methodology	Explore new sources, products, and methods
1.1 Statistical design and build	1.1.2 Design statistics product, 1.1.3 Build statistics product	Develop validated methods into designs and reference implementations; supports all primary capabilities
3.1 Business operations	3.1.1 Personnel management (HR)	Build staff competency in new methods
3.2 Computerisation and automation	3.2.1 Application development	Build reference implementations
4.1 Strategy and leadership	4.1.3 Change management, 4.1.4 Improvement management, 4.1.6 Collaboration	Set the innovation agenda, take in improvement needs, manage the change into production, and cooperate with other NSIs

Stages, processes, and functions

Stage	Realised by processes	Main functions
VS-08.1 Identify Innovation Needs and Opportunities	PRD-1.5 Check data availability and suitability	FUN-4.2 Research and innovate
VS-08.2 Conduct Research and Proof-of-Concepts	No reference process (NSI-specific)	FUN-4.2 Research and innovate
VS-08.3 Evaluate Methodological Soundness and Feasibility	PRD-3.6 Test statistical business process	FUN-4.1 Manage statistical and analytical methodology; FUN-5.2 Assure and review quality
VS-08.4 Develop and Document New Methodologies or Tools	PRD-3.2 Reuse or build processing and analysis components; OVA-5 Knowledge Management	FUN-4.1 Manage statistical and analytical methodology; FUN-10.4 Manage information and records
VS-08.5 Integrate Innovation into Production Environments	PRD-3.7 Finalise production systems	FUN-2.4 Transfer capability improvements to operations; FUN-3.1 Design and build statistical products
VS-08.6 Promote Knowledge Sharing and Capability Building	OVA-5 Knowledge Management	FUN-10.1 Manage human resources; FUN-10.4 Manage information and records

PLACEHOLDER: ArchiMate view VS-08 (b) stages realised by processes and functions

Framework mappings

Cycle step	SAF business processes	GAMSO	SAF capabilities
Plan	PRD-1.1, PRD-1.5, PRD-1.6, PRD-8.3, OVA-1	Strategy and Leadership (set the innovation agenda aligned to the organisational strategy)	2.2.1 Data source exploration, 2.2.5 User research, 4.1.4 Improvement management
Build	PRD-3.2, PRD-3.6, OVA-5, DLM-6	Capability Management (manage research and innovation, develop methodology)	2.2.2 Product innovation, 2.2.3 Methodology, 3.2.1 Application development
Operate	PRD-3.4, PRD-3.7, OVA-2	Capability Management (manage change, transfer to production)	1.1.3 Build statistics product, 4.1.3 Change management
Improve	OVA-5, PRD-8.1 to PRD-8.3	Capability Management (build staff capability)	3.1.1 Personnel management (HR), 4.1.6 Collaboration

7. Principles

Principle	Statement	Application in VS-08
SAF-P01	Actively participate in open-source communities to maintain collective control over statistical processes	Innovate on and contribute to shared, open methods and tools rather than vendor black boxes.
SAF-P05	IT and technology is everyone's business	Connect research, methodology, IT, and business users; build broad capability to adopt new methods.

Principle	Statement	Application in VS-08
SAF-P03	Active Life Cycle management	Manage methods and tools across their life cycle, from proof-of-concept to production to retirement.
SAF-P04	Maximise the benefits for the organisation	Prioritise innovations by organisation-wide value and scalability, not novelty.
SAF-P07	Implement loosely coupled processes and systems	Transfer methods that fit the steady-state interfaces, so that the production streams stay decoupled.

8. Interfaces, services, and events

Business services

Service	Offered to	Realised in	Interfaces
Innovation intake service	VS-01 to VS-06, VS-10	VS-08.1 Identify Innovation Needs and Opportunities	Innovation backlog
Method and tool transfer service	VS-01 to VS-06	VS-08.5 Integrate Innovation into Production Environments	Validated method, reference implementation, design metadata
Knowledge and training service	All staff of the NSI	VS-08.6 Promote Knowledge Sharing and Capability Building	Guidelines, training

Business events

Event	Triggers or marks
Innovation need raised	A primary value stream adds an improvement need to the backlog (VS-08.1)
User-driven innovation need	VS-10 adds a user need that implies a new method (VS-08.1)
Platform request	VS-08 asks VS-09 for a sandbox or production platform
Method compliance assessment	VS-08 submits a method to VS-07 before transfer (VS-08.5)
Production onboarding	The method or tool is onboarded on the VS-09 production runtime
Method transferred to production	Completion of VS-08.5; the method is live in the receiving stream

Interaction with the primary value streams

Counterpart	VS-08 provides	VS-08 receives	Event
All primary streams (VS-01 to VS-06)	Validated methods and tools transferred to production	Improvement recommendations from their *Continuous Improvement* sub-process; fitness-for-purpose validation	*Innovation need raised*, *Method transferred to production*
VS-10 Customer and Stakeholder Engagement	New service and product possibilities	User needs that imply new methods	*User-driven innovation need*
VS-07 Governance and Compliance	Methods for pre-production approval	Compliance criteria and approval	*Method compliance assessment*
VS-09 Infrastructure and Platform Enablement	Requirements for sandbox and production platforms	Research environments and production runtime	*Platform request*, *Production onboarding*

9. Key performance indicators

KPI	Description	Measurement
Idea-to-production lead time	Median time from a backlog item (VS-08.1) to a completed transfer (VS-08.5)	Per transferred innovation
Transfer rate and durability	Methods and tools transferred to production per year, and the share still in use after 12 months	Annually
Gate effectiveness	Share of proofs-of-concept stopped at the VS-08.3 gate with a documented rationale (a healthy gate stops some)	Per evaluation
Backlog responsiveness	Share of prioritised backlog items reaching a proof-of-concept within the agreed time	Per planning period
Capability building	Staff trained on new methods, and adoption rate in the target streams	Per year, per transfer

10. Examples

Smarter imputation. VS-03 reports through its *Continuous Improvement* sub-process that manual follow-up of missing values is too costly. VS-08 tests an imputation method based on machine learning on an authorised copy of State #3, evaluates its accuracy at the VS-08.3 gate, and documents it with a reference implementation. After VS-07 approval it is transferred into the design of VS-03.

New data source. A data space offers a new source for an existing statistic. VS-08 explores the source under specific authorisation, tests whether it can replace part of a survey, and hands the validated connection and method over to VS-01, which then acquires the source in production.

Toolkit: SAF-TK-V08 VS-08 maturity checklist and scenarios